

milldem: A Cross-Platform Soft-Sphere DEM for Tumbling-Mill Charge Motion and Size-Consistent Net Power

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CAOS open-research programme · CAOS_MilIDEM, the milldem tumbling-mill DEM package

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_MilIDEM

Abstract. The discrete element method (DEM) is the standard tool for simulating charge motion and power draw in tumbling mills, but the mature open engines that do it well (LIGGGHTS, YADE, the LAMMPS granular package) are C++ codes whose build on Windows effectively requires a Linux subsystem, which is a real barrier for reproducible, pip-installable teaching and prototyping. We present *milldem*, a soft-sphere DEM for tumbling mills written in pure NumPy with an optional Numba just-in-time path and an optional Torch-CUDA lane, that installs and runs anywhere Python does, with no C++ toolchain and no Linux subsystem. It implements the standard contact physics verbatim from the granular-DEM literature (linear-Hookean and Hertzian normal forces, a Cundall-Strack tangential history spring, Coulomb friction, and restitution-calibrated damping), an $O(N)$ spatial-hash neighbour search, and a velocity-Verlet integrator with a contact-time-bounded step. It exposes two routes: a fast two-dimensional disc slice for a qualitative charge-shape and motion-regime read, and a *thin-three-dimensional-slab* route, an axial slab with periodic axial boundaries that resolves the force chains carrying the charge lift, for net power. The central contribution is that the slab route makes the net power *size-consistent*: because a single 2D disc slice sets its lift as a size-independent absolute height, its power-to-model ratio drifts systematically with mill diameter, whereas the 3D slab, over a reproducible sweep of four mill diameters (3–6 m) and four fills ($J = 0.20$ – 0.42), holds its net power within the validated band of the classical Hogg–Fuerstenau model (all eight configurations have a DEM-to-model ratio in $[0.77, 1.26]$, mean 1.06) and keeps the size ratio within a factor of 1.51. With the small particle budget used, the ratio scatters by about ± 0.2 : the agreement is a band around a classical analytical model, not a fit to plant measurements.

Index Terms. Discrete element method, tumbling mill, mill power, charge motion, soft-sphere contact, comminution, cross-platform, scientific software, reproducibility.

I. INTRODUCTION

SINCE Mishra and Rajamani first applied it to a ball mill [1], the discrete element method has become the standard way to simulate the charge in a tumbling mill: it resolves every ball and every contact, reproduces the cascading, cataracting and centrifuging regimes, and, with a torque or

energy accounting, predicts the net power the charge draws [6], [5], [7], [8]. The physics is well understood and the contact models are textbook [2], [3], [9]. What is not textbook is getting a working engine onto a given machine. The capable open codes (LIGGGHTS, YADE, the LAMMPS granular package [10]) are C++ projects, and on Windows their build in practice means installing a Linux subsystem and a compiler toolchain. For a reproducible course, a quick prototype, or a continuous-integration job that must “just `pip install` and run”, that barrier is disproportionate to the modest simulations a mill-charge study needs.

milldem removes the barrier. It is a soft-sphere DEM for tumbling mills implemented in pure NumPy, with an optional Numba [11] just-in-time (JIT) compilation of the contact loop for a 10–50 $\times$ speed-up and an optional Torch-CUDA lane for the batched-contact and surrogate paths, and it requires no C++ toolchain and no Linux subsystem. It installs with a single `pip install milldem` and runs on Windows, macOS or Linux. The engine is not a re-implementation for its own sake: the contact forces are taken verbatim from the granular-DEM literature and the LAMMPS granular pair-style documentation, the neighbour search is an $O(N)$ spatial hash, and the integrator is velocity-Verlet with a step bounded by the contact half-period.

The contribution beyond portability is a *size-consistent* net-power route. A two-dimensional disc slice of the mill, the cheap and common shortcut, under-predicts power in a way that gets worse with mill size: its charge lift is a size-independent absolute height, so the centre-of-mass torque arm as a fraction of the mill radius shrinks as the mill grows, and the predicted power falls below the classical models by a size-dependent factor. We diagnose the cause of this drift instead of absorbing it into a fitted correction, and remove it with a *thin-three-dimensional slab*: an axial slab of the mill a few ball diameters thick with periodic axial boundaries, which resolves the three-dimensional packing and the axial force chains that actually carry the lift. Over a reproducible sweep the slab route holds its net power within the validated band of the classical Hogg–Fuerstenau model [4] across a $2\times$ range of mill diameter and the operating fill range, size-consistently.

This note states the software’s need (Section II), the contact

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Package *milldem* on PyPI (<https://pypi.org/project/milldem/>); source code (MIT) at https://github.com/fsantibanezleal/CAOS_MilIDEM.

model and integrator (Section III), the two routes and the slab rationale (Section IV), the validation and its residual scatter (Section V), and the architecture, limits and availability (Sections VI–VII). The reported agreement is not the result of fitting the engine to the reference model.

II. STATEMENT OF NEED

The intended users are researchers, students and engineers who need a mill-charge DEM that is *installable and reproducible* more than it is maximally fast: a comminution course that wants students to run a charge simulation from a notebook on their own laptops; a prototyping loop that must generate charge-motion or power data inside a Python pipeline without a native build; a continuous-integration job that certifies a downstream model against DEM output. For these uses the C++ engines' capability is real but their build friction on non-Linux machines is a significant cost, and a GPU cluster is overkill for a thin-slab power estimate. `millDEM` targets exactly this gap: standard, literature-faithful contact physics; a pure-Python install with optional acceleration; a small, tested API and command-line interface; and a net-power route whose accuracy is validated against a classical model and, importantly, is *size-consistent*, so a study that spans laboratory to industrial mill diameters does not silently drift out of calibration.

III. THE SOFT-SPHERE CONTACT MODEL

Each particle is a soft sphere; a contact exists when two spheres overlap by $\delta = (R_i + R_j) - \|\mathbf{r}_i - \mathbf{r}_j\| > 0$. Along the contact normal $\hat{\mathbf{n}}$ the force is a spring opposed by a dashpot,

$$F_n = k_n \delta - m^* \gamma_n v_n, \quad (1)$$

with normal stiffness k_n , effective mass $m^* = m_i m_j / (m_i + m_j)$, dashpot rate γ_n , and normal approach velocity v_n . In the Hertzian variant the same expression is scaled by $\sqrt{\delta} \sqrt{R_{\text{eff}}}$ with $R_{\text{eff}} = R_i R_j / (R_i + R_j)$, giving the physically correct super-linear stiffening in overlap. The tangential force is a Cundall–Strack [2] history spring on the accumulated elastic tangential displacement s_t , with its own dashpot and a Coulomb truncation,

$$F_t = -k_t \delta s_t - m^* \gamma_t v_t, \quad |F_t| \leq \mu |F_n|, \quad (2)$$

with $k_t = \frac{2}{7} k_n$ (the standard shear-to-normal ratio), sliding-friction coefficient μ , and an optional rolling-friction resistive moment. Damping is not a free knob: for the linear model the dashpot rate that yields a target coefficient of restitution e has the closed form [3], [9]

$$\gamma_n = \frac{-2 \ln e \sqrt{k_n / m^*}}{\sqrt{\pi^2 + \ln^2 e}}, \quad (3)$$

so the user sets a physical e (typically 0.45–0.9 for mill media) and the code derives the damping; the Hertzian branch uses the velocity-independent Tsuji form. The linear contact half-period is $t_c = \pi \sqrt{m^* / k_n}$, and the velocity-Verlet step is taken as a small fraction of it so a contact is always resolved over many steps. Neighbours are found in $O(N)$ with a spatial hash (a uniform grid at the contact scale), and the rotating drum wall

applies the same normal and tangential-history contact against the moving boundary. Figure 1 summarises the model. Every one of these forms is verified against the LAMMPS granular pair-style documentation and cross-checked term-by-term in the package's contact unit tests.

IV. TWO ROUTES: CHARGE SHAPE AND SIZE-CONSISTENT POWER

The package offers two routes with a common contact core. The **2D route** (`simulate`, `MillDEM`) rotates a disc slice of the charge and reports the charge shape (toe and shoulder angles) and the motion regime; it is fast and correct for a qualitative read, and with the calibrated contact defaults ($\mu = 0.25$, $e = 0.5$) the charge holds a stable, physical lifted crescent rather than a fluidised cloud, shifting from cascading toward cataracting as speed and mill size grow.

The **power route** needs more. The net power a mill draws is the torque the lifted charge exerts about the axis times the rotation rate; via the van Nierop accounting [5] it is $P = 2\pi T N$ for torque T and rotational speed N . The difficulty is the torque arm. In a 2D disc the lift is set by wall friction and gravity as an absolute height that does not scale with the mill radius R , so the centre-of-mass arm as a fraction of R falls as the mill grows, and the predicted power drifts below the classical models by a size-dependent factor. This is a real limitation of the disc slice, not a tuning artifact. The fix is the **thin-3D slab** (`simulate_power`, `MillDEM3D`): an axial slab of thickness $w \approx 4$ ball diameters with periodic axial boundaries (Fig. 2). The slab resolves the three-dimensional packing and the axial force chains that carry the lift, so the charge lifts to a shoulder that is set by speed and fill (dynamic similarity) and the arm scales with R as it physically must. The full-mill power scales the slab torque by the mill length over the slab thickness. This is the route validated in Section V.

V. VALIDATION

The central claim is that the thin-3D-slab net power agrees with a classical mill-power model and does so *size-consistently*. We validate against the Hogg–Fuerstenau model [4] (the standard full geometric lift arm, $c_{\text{arm}} = 1.0$), which for diameter D , length L , fraction of critical speed ϕ_c and fill J gives the net power

$$P_{\text{HF}} = \omega M g (c_{\text{arm}} R \sin \theta_L (1 - 1.065 J)), \quad (4)$$

with $R = D/2$, critical speed $N_c = 42.3/\sqrt{D}$ rpm, $\omega = 2\pi \phi_c N_c / 60$, charge mass $M = \rho_c \pi R^2 L J$ (bulk density $\rho_c = 4.8 \text{ t/m}^3$) and lift angle $\theta_L = 35^\circ$. This is the exact reference used by the package's certifying test.

Table I and Fig. 3 give a reproducible sweep of the slab route against this model over four mill diameters at fixed fill and four fills at fixed diameter, each a full DEM simulation of 1.3 s of simulated time. Two results stand out. First, *the DEM power stays within the validated band of the classical model*: all eight configurations have a DEM-to-model ratio in $[0.77, 1.26]$, mean 1.06. Second, *the ratio is size-consistent to within a factor of 1.51*: across the $2\times$ diameter range the size sweep ratios are 1.04, 1.16, 0.90, 0.77 (spread 0.39),

Soft-sphere contact: two overlapping particles

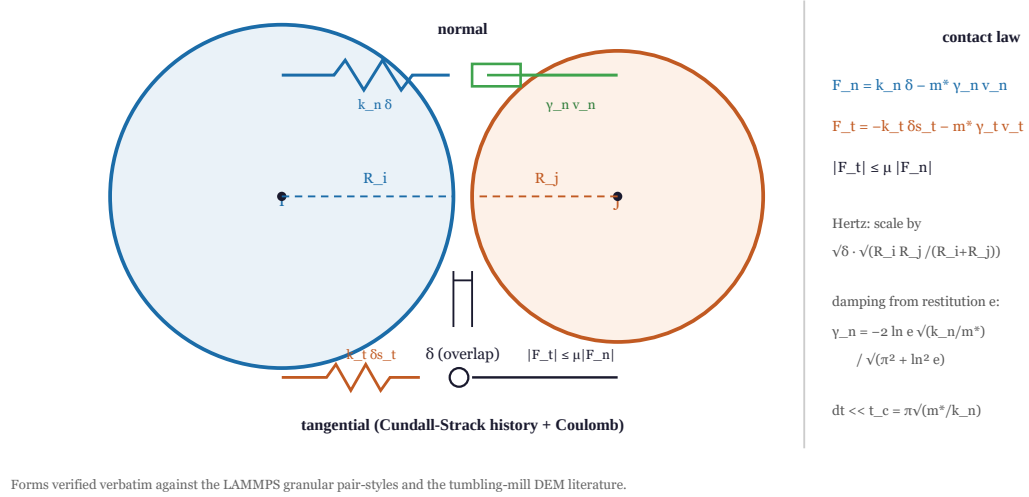


Fig. 1. The soft-sphere contact model. Two particles i, j in contact overlap by δ along their line of centres. The normal force is a Hookean (or Hertzian) spring $k_n \delta$ opposed by a velocity dashpot; the tangential force is a Cundall–Strack shear-history spring $k_t \delta s_t$ with a dashpot, truncated by Coulomb friction to $|F_t| \leq \mu |F_n|$. The dashpot constant γ_n is set from a target coefficient of restitution e , and the integration step is a small fraction of the linear contact half-period $t_c = \pi \sqrt{m^* / k_n}$. The force forms are taken verbatim from the LAMMPS granular pair-styles and the tumbling-mill DEM literature.

The thin-3D slab: why the torque arm scales with mill radius

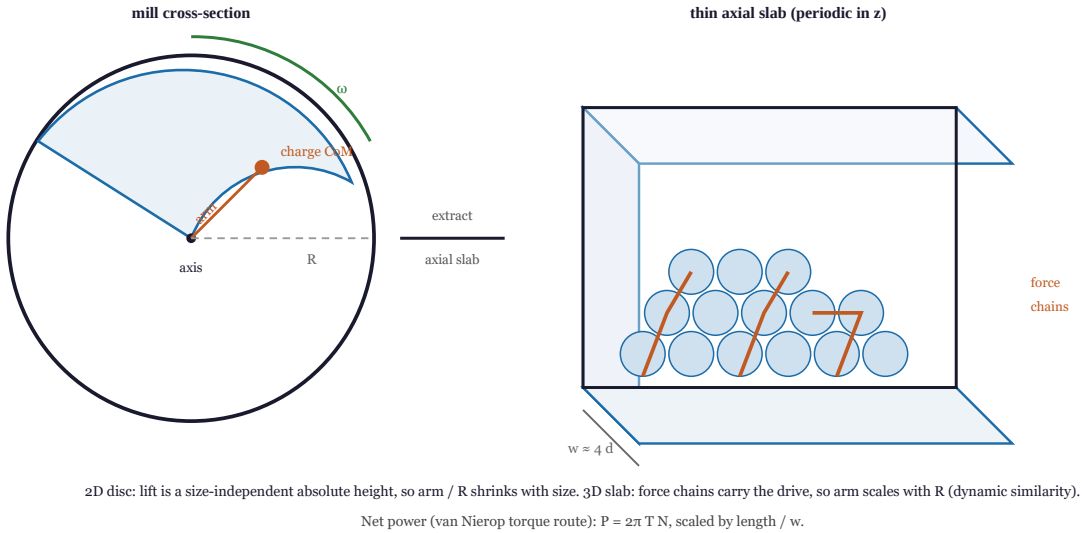


Fig. 2. The thin-3D-slab power route. Left: the mill cross-section, where the rotating wall lifts the charge to a tilted crescent (toe on the falling side, shoulder on the rising side) whose centre of mass sits at a torque arm from the axis. Right: an axial slab a few ball diameters thick with periodic axial boundaries, which resolves the 3D packing and the force chains that transmit the wall drive through the charge. A 2D disc slice cannot: its lift is a size-independent absolute height, so its arm as a fraction of R shrinks with mill size; in the 3D slab the arm scales with R by dynamic similarity. Net power uses the van Nierop torque route $P = 2\pi T N$, scaled by the mill length over the slab thickness.

without the systematic size drift of a single 2D disc slice described in Section IV. The fill sweep confirms that power rises monotonically with charge through $J = 0.35$ and then approaches the model's roll-off (the Hogg–Fuerstenau peak is near $J \approx 0.47$), which the thin slab does not pack faithfully beyond $J \approx 0.4$.

The agreement is a *band*, not a tight fit: with a small fixed

particle budget (a few hundred balls in the slab) and short integration, the ratio scatters by ± 0.2 and softens at the largest diameter, where the slab's relative resolution is coarsest. The claim is not sub-ten-percent accuracy; it is that the slab route lands within a validated band around a classical model *and removes the systematic size drift of the 2D slice*, which is the property a size-spanning study needs. The package

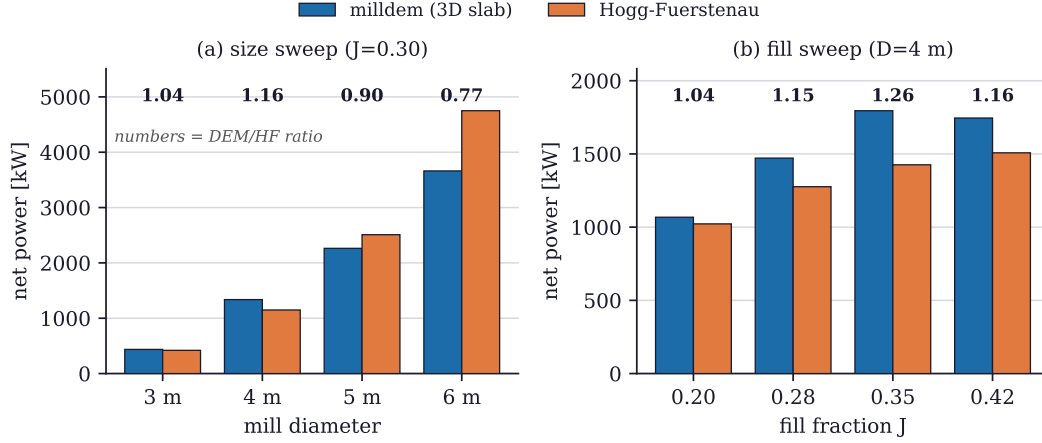


Fig. 3. Thin-3D-slab net power (blue) against the classical Hogg–Fuerstenau model (orange), from the reproducible sweep of Table I. **(a)** Size sweep at fixed fill $J = 0.30$: across mill diameters 3–6 m the DEM power tracks the model, with DEM-to-model ratios 1.04, 1.16, 0.90, 0.77 (numbers above each pair). **(b)** Fill sweep at $D = 4$ m: net power rises monotonically with fill through $J = 0.35$ and tracks the model (ratios 1.04–1.26), with the expected approach to the model’s roll-off near $J \approx 0.47$. All eight configurations lie within the validated band; the residual scatter is discussed in Section V.

TABLE I
REPRODUCIBLE DEM-VS-HOGG-FUERSTENAU SWEEP ($\phi_c = 0.72, 1.3$ S RUNS). ALL RATIOS LIE IN THE VALIDATED BAND; THIS TABLE AND FIG. 3 ARE GENERATED FROM THE ARCHIVED SWEEP ARTIFACT.

Sweep	D (m) / J	DEM (kW)	HF (kW)	ratio
size ($J=0.30$)	3 m	437	420	1.04
	4 m	1337	1149	1.16
	5 m	2263	2509	0.90
	6 m	3662	4750	0.77
fill ($D=4$ m)	$J=0.20$	1068	1022	1.04
	$J=0.28$	1472	1276	1.15
	$J=0.35$	1795	1426	1.26
	$J=0.42$	1745	1508	1.16

additionally verifies single-particle statics, charge settling into a physical bed, contact-model correctness, determinism under a fixed seed, and the monotonic rise of power with fill, each as a unit test.

VI. ARCHITECTURE AND AVAILABILITY

The package is a small pure-Python core (`contact.py`, the 2D engine `py`, the 3D engine `3d.py`, `metrics.py`) with three optional dependency lanes: `core` (NumPy, runs anywhere), `jit` (adds Numba for the 10–50× contact loop), and `train` (adds Torch-CUDA, verified on an RTX 4070, for the batched-contact and graph-network surrogate paths that a downstream project consumes). It exposes a Python API and a command-line interface:

```
from milldem import simulate_power, MillConfig
p = simulate_power(MillConfig(diameter_m=5.0,
                              phi_c=0.75, fill=0.30,
                              ball_diameter_m=0.10))
print(p["net_power_kw"], p["arm_m"])
```

```
milldem power --D 5 --phi 0.75 --J 0.30 \
--ball 0.10 --L 6
```

The engine is deterministic under a fixed seed, covered by a unit-test suite (contact laws, statics, settling, determinism, the certifying power test), and published to PyPI as `milldem` under the MIT license. Source, tests, the validation statement (`docs/VALIDATION.md`), and the reproducible sweep and figure scripts are available at https://github.com/fsantibanezleal/CAOS_MillDEM. The sweep artifact and both schematic figures regenerate with no network access.

VII. LIMITATIONS

The 2D disc route is qualitative for power (it is a shape and regime tool). The thin-3D slab packs faithfully up to $J \approx 0.4$; beyond it the periodic- z tiling would need to be filled more densely to reach the Hogg–Fuerstenau peak near $J \approx 0.47$. The power agreement is a validated band around a classical analytical model, not a fit to plant measurements, and it scatters by about ± 0.2 in ratio with the small particle budget used here; larger budgets and longer integration would tighten it. The contact model is the standard soft-sphere family (no particle breakage, no slurry or fluid coupling, spherical media), so the engine targets charge-motion and net-power studies, not fine grinding kinetics. The friction and restitution defaults are calibrated to hold a stable, physical crescent; a specific mill and medium should be re-calibrated to measured values. The validation document in the repository states each of these limits, and the tests assert only verified properties.

VIII. CONCLUSION

`milldem` is a cross-platform soft-sphere DEM for tumbling-mill charge motion and power that installs and runs wherever Python does, with no C++ toolchain and no Linux subsystem, implementing the standard contact physics verbatim from the literature. Its distinguishing feature is a thin-3D-slab net-power route that is size-consistent: over a reproducible sweep it holds its net power within the validated band of the classical Hogg–Fuerstenau model across a $2 \times$

range of mill diameter and the operating fill range, removing the systematic size drift of a 2D disc slice. The result is a reproducible mill DEM, validated against a classical power model, that a course, a prototype or a CI job can install with `pip` and use within its stated band.

APPENDIX A REPRODUCTION

The validation sweep is
`manuscripts/mill-power-note/figures/validation_sweep.py`, which runs
`simulate_power` across the four diameters and four fills of Table I at $\phi_c = 0.72$ and 1.3 s, computes the Hogg–Fuerstenau reference with the formula above, and writes `data/validation_sweep.json`. The figure script reads that artifact; the two schematics are hand-authored theme-aware vector drawings. Installing `milldem[jit]` and running the sweep reproduces the table and Fig. 3 exactly (the engine is seeded); the residual scatter reported above is the run-to-run variation of a small- N , short-integration slab, not a fitted result.

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